
Technical Deliverable

Client 1: Technical Codebook

Refreshed PSID crisis-module workflow, scoring logic, and artifact schema

Document status

Prepared for client review

Updated: April 2026

Repository: Team-PSID

Contents

1	Purpose and Scope	1
2	Headline Benchmark Summary	2
3	Why 28 Scored Rows Become 26 Deployable Questions	3
4	Figures and Visual Diagnostics	4
5	End-to-End Workflow Pipeline	6
6	Optimization Strategy Change	7
7	Respondent-Answer Refinement and Optimization	8
8	Files and Roles	9
9	Core Scoring Logic	9
9.1	Utility, burden, and baseline rank	9
9.2	Binary category assignment	10
9.3	Augmented priority score	10
9.4	Why each quantitative field exists	12
10	Worked Calculation Example	13
11	Threshold Diagnostics	15
12	Construct Coverage and Interaction Notes	15
12.1	Selected construct coverage	16
12.2	Demographic interaction summary	16

13 Output Schema	17
14 Deliverables Produced From the Same Dataset	17
15 Refresh Procedure	18

1 Purpose and Scope

This technical codebook documents the refreshed PSID crisis-module workflow from the authoritative ranked source file through the final scored CSV, figures, spreadsheet, and PDF bundle. It is written to support analytic review, reproducibility, and downstream client sign-off.

Production status

- 52 ranked historical questions in the current corpus.
- 28 selected rows in the scored module and 26 questions in the deployable instrument.
- 3 **Generic** rows and 49 **Specific** rows after binary thresholding.
- 29.98 selected minutes in the scored module and 28.82 minutes in the deployable instrument.
- The 28-to-26 change comes from excluding `SPC_AGE_YEARS` and `SPC_DESCRIBE_SEX_GENDER` from routine field administration while retaining them for traceability.

2 Headline Benchmark Summary

Metric	Value	Calculation / rule	Why it is used and the assumption behind it
Total ranked questions	52	$\text{count}(\text{all rows}) = 52$	Starting analytic universe; assumes the integrated source file is the authoritative corpus for each refresh.
Selected scored questions	28	$\text{count}(\text{selected_for_module} = \text{True}) = 28$	Defines the scored review package under the fixed 30-minute module cap.
Deployable questions	26	$28 - 2 = 26$ after removing selected demographic-only traceability rows	Produces the live instrument; assumes demographic context can be collected elsewhere or only when routing requires it.
Selected scored minutes	29.98	$\sum \text{minutes over the 28 selected} = 29.98$	Confirms the scored module stays within the client's review budget.
Deployable minutes	28.82	$29.98 - 0.35 - 0.817 = 28.813 \approx 28.82$	Paper-length estimate for the field instrument after age and sex/gender are removed from routine administration.
Full corpus minutes	68.60	$\sum \text{minutes over all 52 rows} = 68.60$	Shows why a selection layer is required; assumes every row would otherwise be asked.
Generic rows	3	$\text{count}(\text{ri_threshold_score} \geq 0.70) = 3$ after normalization	Keeps the universal portable core intentionally small; assumes the binary split should surface only the strongest cross-crisis items.
Specific rows	49	$52 - 3 = 49$	Keeps direct crisis-detail items in the context-sensitive bank.
Average R_i	2.201	$\text{mean}(R_i) = 2.201$	Benchmark statistic only; not itself a cutoff rule.
Maximum R_i	5.667	$\text{max}(R_i) = 5.667$	Upper bound for min-max normalization in the current refresh.
Average P_i	2.561	$\text{mean}(P_i) = 2.561$	Tracks distinctiveness-adjusted priority across the corpus.
Prepared for client review Maximum P_i	7.860	$\text{max}(P_i) = 7.860$	Shows the ceiling after rarity, construct, portability, and redundancy adjustments.

3 Why 28 Scored Rows Become 26 Deployable Questions

Stage	Rows	Minutes	Calculation / rule	Why it is retained or removed
Full ranked corpus	52	68.60	Count of all rows in the integrated source; minutes are summed across the corpus	Starting point for ranking, diagnostics, and documentation.
Scored module	28	29.98	Count of selected rows after ranking and time-budget filtering	Preserves the full analytic shortlist, including traceability fields.
Selected demographic traceability rows	2	1.17	SPC_AGE_YEARS (0.35) + SPC_DESCRIBE_SEX_GENDER (0.817)	Useful for framing and optional routing, but comparatively weak as direct crisis-impact questions.
Deployable instrument	26	28.82	$28 - 2 = 26$ and $29.98 - 1.167 = 28.813 \approx 28.82$	Final asked instrument prioritizes direct crisis-impact content instead of forcing demographic context into every interview.

Table 2: The change from 28 scored rows to 26 deployable questions is a deployment optimization, not a re-ranking of the corpus.

Interpretation

The model still selects 28 rows as the best analytic bundle under the budget. The deployable version removes only the two demographic-only rows because they are better treated as traceability or conditional-routing variables than as routine crisis-module questions.

4 Figures and Visual Diagnostics

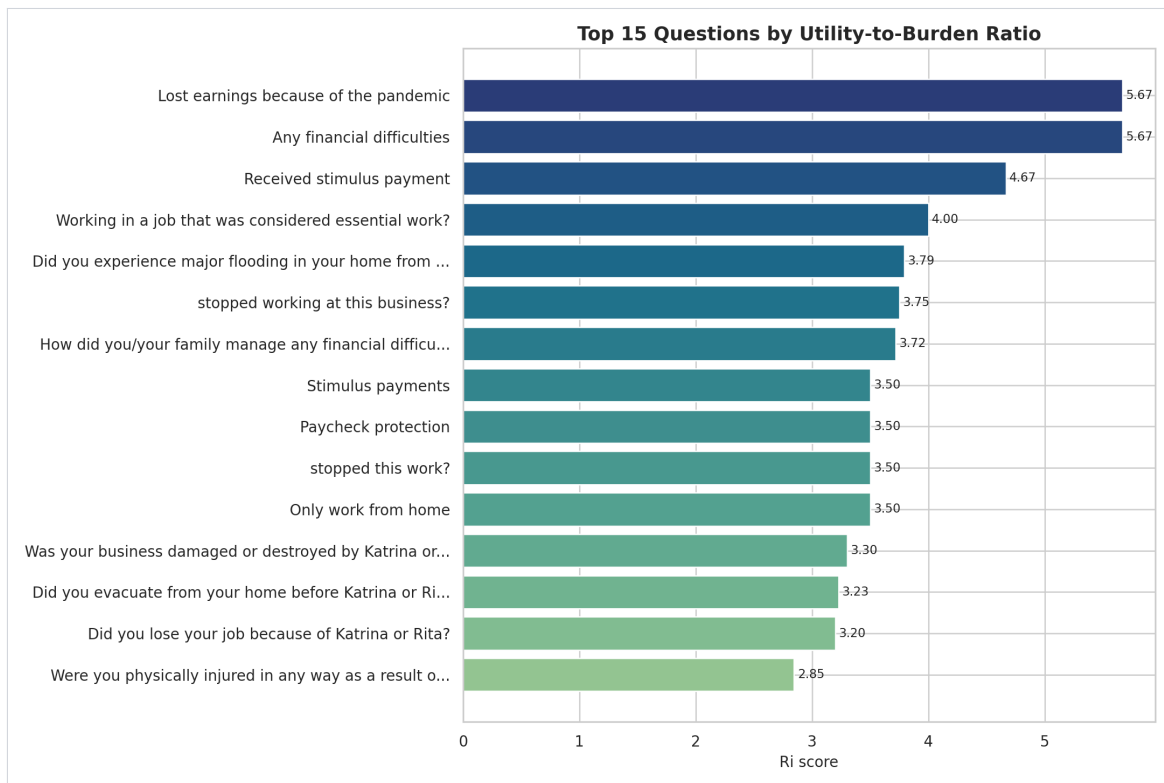


Figure 1: Top-ranked questions from the refreshed corpus.

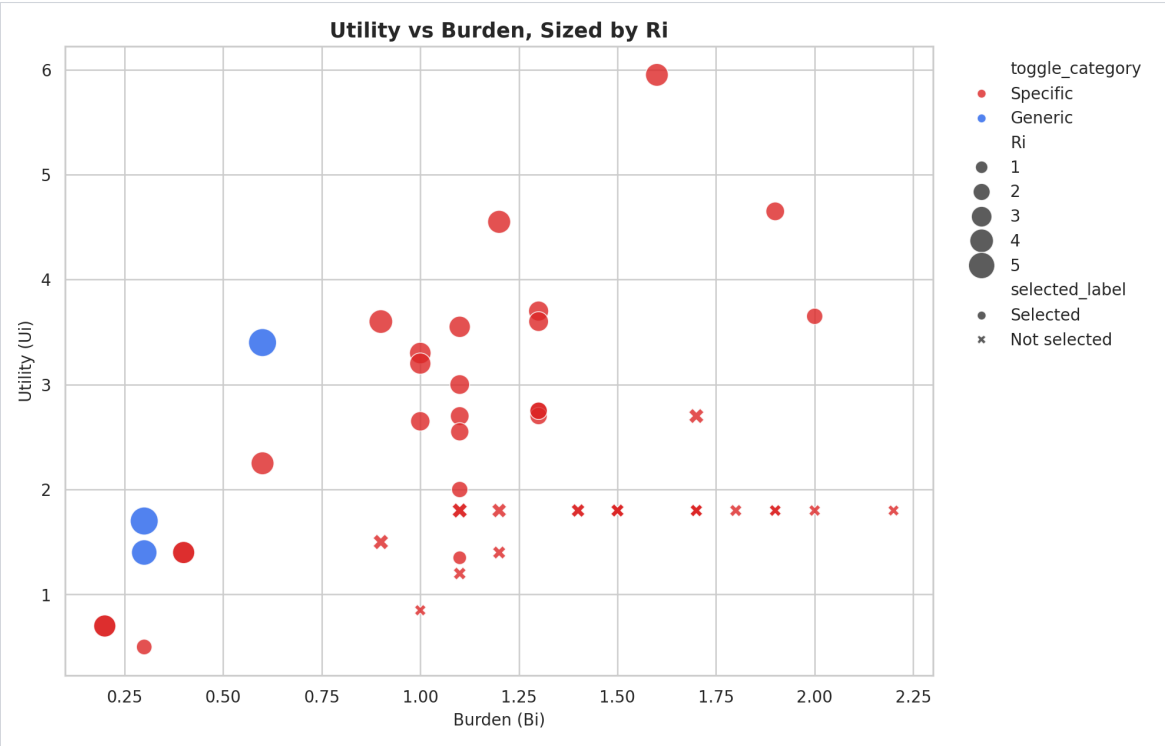


Figure 2: Utility-versus-burden view used to inspect signal efficiency.

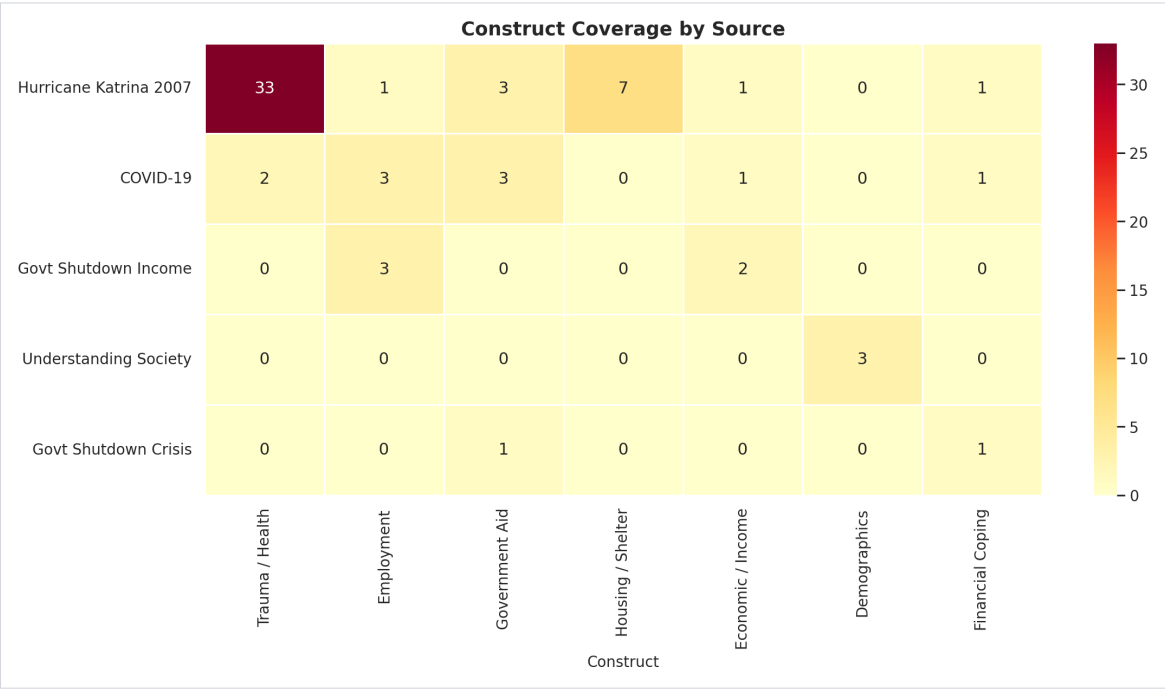


Figure 3: Construct coverage heatmap for the refreshed item bank.

5 End-to-End Workflow Pipeline

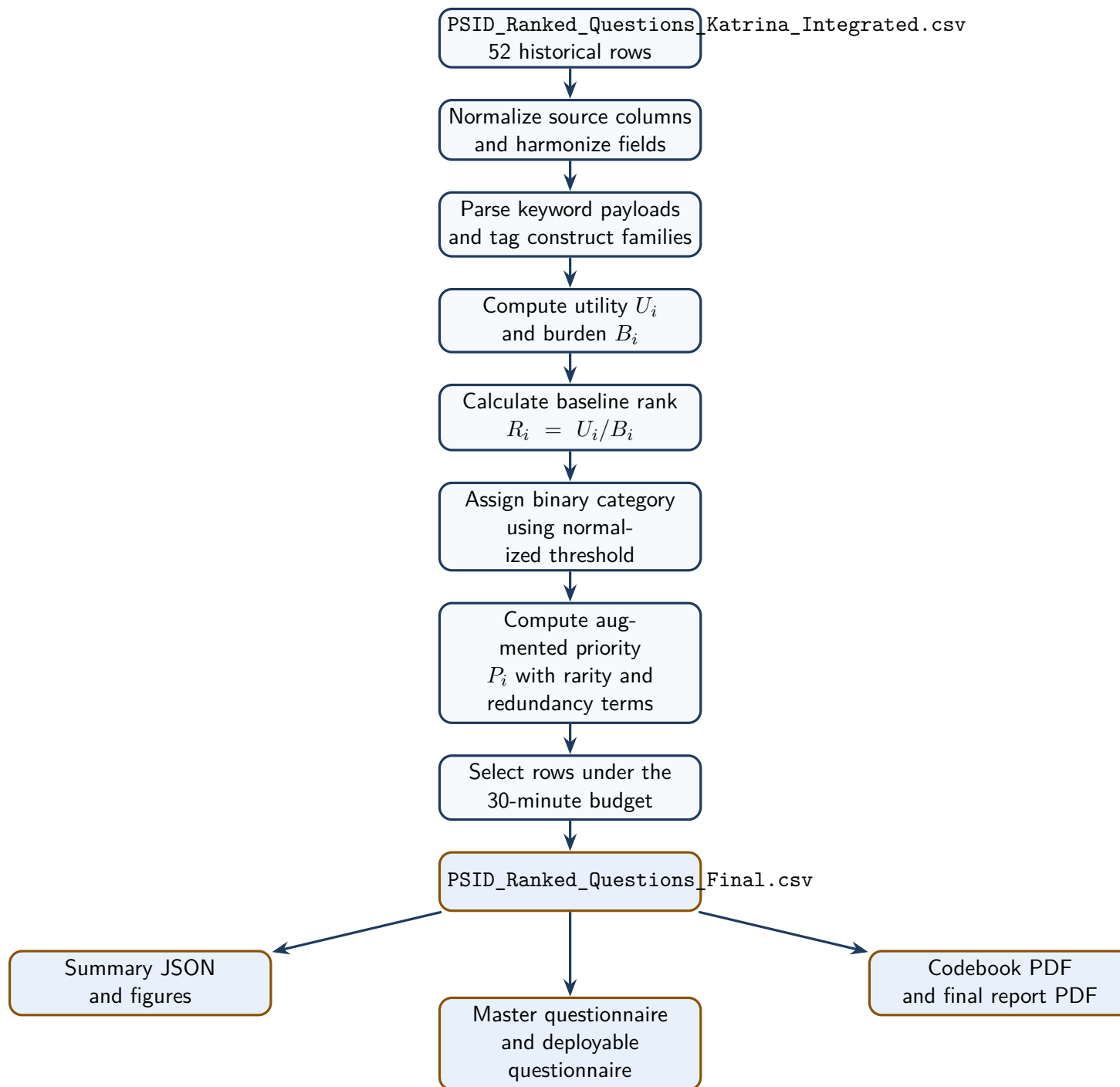


Figure 4: Production workflow from the raw integrated source to the maintained client package.

6 Optimization Strategy Change

Dimension	Legacy architecture view	Current production view	Why the change matters
Client-facing categories	Legacy architecture used 7 Generic Core rows, 1 Financial Crisis row, and 20 Pandemic / Disaster rows inside the same 28-row shortlist	Current production uses 3 Generic and 25 Specific rows in the scored module, with 3 Generic and 23 crisis-specific rows in the deployable instrument	The client narrative is now binary and easier to defend; the universal core is intentionally narrower.
Financial representation	Finance appeared as its own visible toggle label	Financial rows now compete inside Specific and are moderated by redundancy penalties	Near-duplicate earnings and aid items no longer create the impression of a separate finance-only module.
Deployment target	The scored shortlist was closer to the field-facing view	The 28-row scored file is retained for auditability, while the 26-row deployable file is used for routine administration	Separates analytic traceability from operational burden.
Optimization levers	Score plus legacy toggle structure	Score plus portability bonus, construct coverage, redundancy control, and documentation-backed wording replacements	The instrument is optimized for an always-on, cross-crisis deployment setting rather than only for archival structure.
Routing refinement	Category labels described broad source-specific routing	Generic answers and crisis identification activate one relevant specific block instead of treating the full package as a single interview path	Respondent answers now reduce effective burden in live administration without overwriting the stored row scores.

Table 3: From the legacy multi-label architecture to the current production optimization strategy.

7 Respondent-Answer Refinement and Optimization

Refinement input	Quantitative effect	Why it is used	Working assumption
Universal Generic answers and crisis identification	Effective live path becomes the 3-item Generic core plus one specific block; for example pandemic path minutes are $1.40 + 2.92 = 4.32$ instead of all 26 deployable items	Cuts unnecessary burden and keeps the interview aligned to the active crisis context	One dominant crisis context is active in a given administration.
Selected demographic answers	Removing SPC_AGE_YEARS (0.35) and SPC_DESCRIBE_SEX_GENDER (0.817) lowers the base deployable length by 1.167 minutes	Preserves traceability without forcing low-yield context items into every interview	Demographic information can be captured elsewhere or only when routing requires it.
Historical respondent distributions	Shutdown coping, stimulus uptake, and pandemic response patterns support keeping Government Aid and Financial Coping constructs salient in the selected bank	Grounds construct weights and wording choices in observed crisis behavior rather than only lexical similarity	Historical respondent patterns are informative for future crisis-module design.
Feed-forward respondent state	Prior-wave status, identity, address, and eligibility can refine universes or wording windows, but do not overwrite stored U_i , B_i , R_i , or P_i values	Distinguishes static ranking from live survey routing	Prior-wave metadata is available when the module is deployed longitudinally.

Table 4: Respondent data influences deployment and construct emphasis even when the stored row scores remain fixed.

The respondent-data evidence used for refinement is substantive rather than cosmetic. In the reference evidence, 70.6% of shutdown-affected families cut spending, 28.3% used food banks, and pandemic response data show strong heterogeneity in vaccination and stimulus

behavior. Those patterns explain why coping, aid, earnings-loss, and health-response items remain prominent after redundant variants are pruned.

8 Files and Roles

File	Role	Why it matters
data/PSID_Ranked_Questions_Kinr Integrated Starting	Integrated source bank	Starting point for every production rebuild.
PSID_NLP_Crisis_Module_Structure_helpers	Structure copying helpers	Holds constants, keyword parsing, burden logic, threshold helpers, and selection utilities.
generate_psid_artifacts.py	Production rebuild engine	Writes the final scored CSV, summary JSON, dashboard payload, and figures.
data/PSID_Ranked_Questions_Final scored output	As Final table	Main ranked table used by the notebook and client deliverables.
PSID_NLP_Crisis_Module_Final validation notebook	Validation notebook	Recomputes the workflow and verifies the persisted artifact bundle.
generate_client_documents.py	Client-package renderer	Builds the spreadsheet and PDFs from the scored CSV and summary JSON.
data/psid_artifact_summary.json	Artifact bundle	Stores benchmark counts and score summaries for the latest run.

Table 5: Files that drive the maintained workflow and client package.

9 Core Scoring Logic

9.1 Utility, burden, and baseline rank

The baseline score is driven by question utility relative to response burden.

Core formulae

$$U_i = \sum \text{keyword weights}$$

$$B_i = \max(0.10 \times \text{word_count} + 0.20 \times \text{complexity}, 0.1)$$

$$R_i = \frac{U_i}{B_i}$$

- U_i rewards crisis-relevant content signaled by tagged keywords.
- B_i penalizes long or complex wording.
- R_i ranks items that carry more signal per unit of burden.

9.2 Binary category assignment

The client requirement was a binary rule based on R_i , but the raw score is not naturally bounded in $[0, 1]$. The production pipeline therefore applies the cutoff to a min-max normalized score.

Operational threshold rule

$$\text{ri_threshold_score} = \frac{R_i - \min(R_i)}{\max(R_i) - \min(R_i)}$$

$$\text{Generic} \iff \text{ri_threshold_score} \geq 0.70$$

$$\text{Specific} \iff \text{ri_threshold_score} < 0.70$$

9.3 Augmented priority score

The final prioritization score preserves R_i while rewarding breadth and distinctiveness.

Priority adjustment

$$\begin{aligned} \text{augmented_utility} &= U_i \times \left(1 + 0.32 \times \text{idf_scaled} + 0.24 \times \text{construct_scaled} \right. \\ &\quad \left. + 0.12 \times \text{richness_scaled} + \text{portability_bonus} \right) \\ P_i &= \frac{\text{augmented_utility}}{B_i \times (1 + 0.65 \times \text{redundancy_scaled})} \end{aligned}$$

- P_i helps separate distinct, portable items from repetitive variants.
- The shortlist is still bounded by the time budget, not only by raw rank.

9.4 Why each quantitative field exists

Quantity	Calculation	Why it is used	Assumption
U_i	$\sum$ keyword weights	Captures substantive crisis relevance signaled by tagged constructs	Tagged keywords are a defensible proxy for item content.
B_i	$\max(0.10 \times \text{word_count} + 0.20 \times \text{complexity}, 0.1)$	Penalizes long or cognitively difficult wording	Administration burden grows approximately with length and complexity.
R_i	U_i/B_i	Measures signal per unit of burden	High-value questions should be informative relative to the time they consume.
$\text{ri_threshold_score}$	$(R_i - \min R_i) / (\max R_i - \min R_i)$	Makes the client's 0.70 threshold meaningful on an otherwise unbounded score	The current corpus minimum and maximum are authoritative for the refresh.
augmented utility	$U_i \times (1 + 0.32 \text{idf_scaled} + 0.24 \text{construct_scaled} + 0.12 \text{richness_scaled} + \text{portability_bonus})$	Rewards rarity, coverage breadth, and wording portability before redundancy is applied	Distinct and portable content adds more deployment value than repeated variants.
P_i	$\text{augmented_utility} / [B_i \times (1 + 0.65 \text{redundancy_scaled})]$	Breaks ties among similar R_i values by down-weighting repetitive items	Short modules should prefer distinct questions over clusters of near-duplicates.
minutes	$(\text{word_count} \times 7) / 60$	Operational time field used in the budget and instrument assembly	Oral administration takes about 7 seconds per word on average.
selected rows	Time-budget filter after ranking	Defines the scored module for review and traceability	The 30-minute scored-module cap is fixed for the current package.
deployable rows	Selected rows minus demographic-only traceability rows	Defines the field instrument	Demographic context is not part of the core crisis ask unless routing requires it.

Table 6: Formula, purpose, and assumption guide for the main quantitative fields.

10 Worked Calculation Example

Example question: GEN_LOSE_EARNINGS_PANDEMIC

Recommended wording: *Did you lose earnings because of the pandemic?*

Field	Value	Calculation	Why it is used / assumption
U_i	3.400	$0.80 + 0.80 + 0.90 + 0.90$	Utility aggregates the matched crisis-signal weights; assumes the tagged keywords capture the substantive content of the item.
word count	6	Counted from deployable wording	Used in both burden and time estimates; assumes the cleaned wording is the administered string.
complexity	0.0	Precomputed complexity field	Prevents an additional burden penalty; assumes the wording is direct and does not require extra respondent processing.
B_i	0.600	$\max(0.10 \times 6 + 0.20 \times 0.0, 0.1)$	Converts wording cost into a penalty term; assumes the burden constants remain fixed design choices.
R_i	5.667	$3.40/0.60$	Measures signal efficiency; assumes high-value universal questions should deliver strong utility per unit of burden.
threshold score	1.000	$(5.667 - 0.818)/(5.667 - 0.818)$	Supports the binary split; assumes the current refresh min and max are the correct normalization anchors.
Final category	Generic	$1.000 \geq 0.70$	Places the item in the universal core; assumes the client wants only the top portable items to be Generic.
P_i	7.860	Value computed after rarity, construct, richness, portability, and redundancy scaling	Separates this item from similar crisis questions; assumes distinct portable items deserve priority in a short module.
minutes	0.70	$(6 \times 7)/60$	Feeds the time budget; assumes oral administration averages 7 seconds per word.

Table 7: Worked example for one of the highest-value Generic items.

11 Threshold Diagnostics

Threshold diagnostic	Value	Calculation / implication
Minimum raw R_i	0.818	$\min(R_i)$ across the 52-row corpus.
Maximum raw R_i	5.667	$\max(R_i)$ across the 52-row corpus.
Rows with raw $R_i \geq 0.70$	52 of 52	Shows that a direct raw cutoff would classify every row as Generic.
Rows with normalized threshold score ≥ 0.70	3 of 52	Final Generic count after min-max normalization.

Table 8: Why normalized thresholding is required for a meaningful binary split.

Interpretation

Raw $R_i \geq 0.70$ would classify every row as **Generic**. Normalized thresholding preserves the requested cutoff while still creating a real binary split between portable high-signal items and context-dependent items.

12 Construct Coverage and Interaction Notes

12.1 Selected construct coverage

Construct	Selected ques- tions
Trauma / Health	15
Employment	7
Housing / Shelter	6
Government Aid	5
Economic / Income	4
Financial Coping	2
Demographics	2

Table 9: Construct coverage in the selected scored module.

12.2 Demographic interaction summary

Group	Rows	Average R_i	Average P_i
Demographics only	3	1.354	2.189
Non-demographic crisis items	49	2.253	2.584

Table 10: Demographic-only rows versus direct crisis-impact rows.

Interpretation

Demographic-only items remain useful for traceability and module framing, but they carry less direct crisis signal than employment, housing, aid, trauma, and economic disruption items. That is why the final deployable questionnaire excludes demographics even though the master review file retains them.

13 Output Schema

Field	Meaning
variable_name	Stable identifier used across the CSV, spreadsheet, and PDFs.
question_text	Historical source wording.
recommended_wording	Clean deployable wording.
source	Historical module source.
toggle_category	Final binary category: Generic or Specific .
historical_toggle_category	Legacy category kept for traceability.
U_i, B_i, R_i	Utility, burden, and baseline rank.
ri_threshold_score	Normalized rank used for binary categorization.
P_i	Enhanced priority score.
selected_for_module	Final selection flag under the time cap.
minutes	Estimated administration time.
constructs	Tagged construct families.

Table 11: Key fields preserved in the authoritative scored dataset.

14 Deliverables Produced From the Same Dataset

Deliverable	Audience	Purpose
deliverables/Master_Questionnaire.xlsx	Internal review	Full 52-row review sheet with categories, wording, and scores.
deliverables/Deployable_Questionnaire.pdf	Survey deployment	Final question text without demographic rows.
deliverables/Codebook.pdf	Technical review	Rendered technical codebook.
deliverables/Final_Report.pdf	Stakeholder review	Narrative summary of results and interpretation.
deliverables/Client_1_Technical_Codebook.pdf	Client circulation	LaTeX report version of this technical codebook.

Table 12: Client-facing and internal deliverables generated from the same scored dataset.

15 Refresh Procedure

Use this order whenever the client package is refreshed:

1. Rebuild scored artifacts via `build_all()` in `generate_psid_artifacts.py`.
2. Validate notebook outputs in `PSID_NLP_Crisis_Module_Final.ipynb`.
3. Regenerate spreadsheet and PDFs through `generate_client_documents.py`.
4. Rebuild the client-facing LaTeX reports through `docs/latex/build_client_reports.sh`.
5. Confirm that metrics, figures, tables, and deliverable paths remain aligned.

That sequence keeps the scored CSV, the notebook, the figures, the packaged PDFs, and the presentation-ready report files synchronized.